

Experience Sharing Box: A Platform for Capturing, Storing, and Playback of Multisensory Memory

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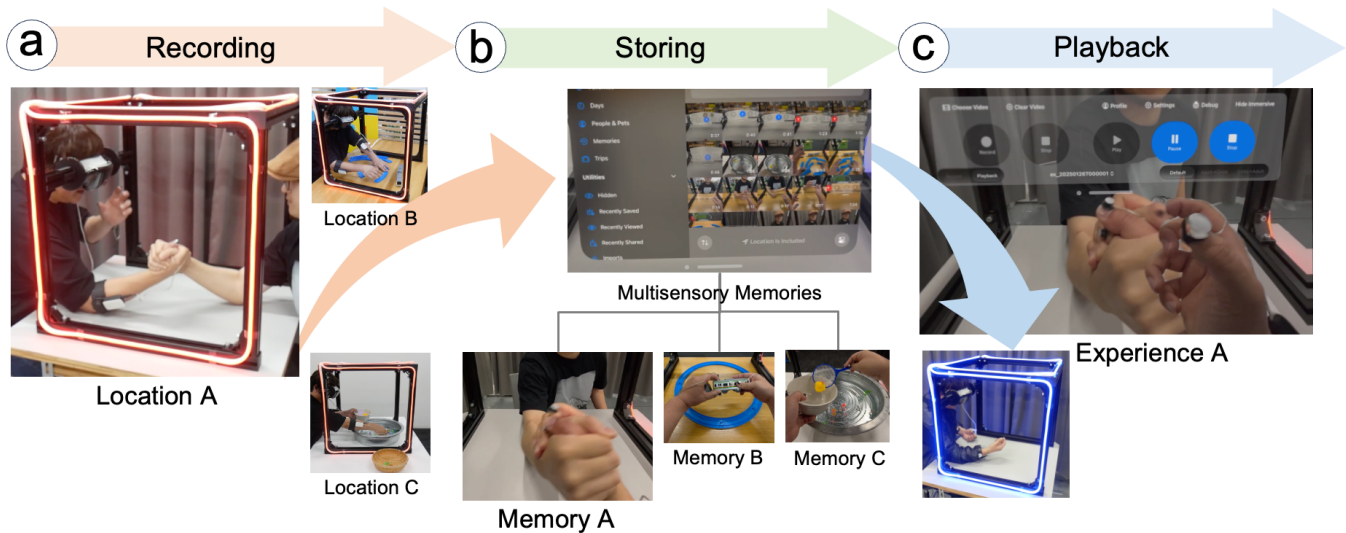


Figure 1: Overview of the Experience Sharing Box (ESB) and Experium Platform. The ESB comprises a spatial display rigidly mounted inside the cube-shaped box and Haptic wearable devices. All hardware components are orchestrated by the Experium Platform, a software stack for recording, storing, and playback Multisensory Memories. (a) Users can capture Multisensory Memories at any location and at any moment. (b) Each memory is linked to a unique identifier and stored in a distributed manner across device-local and cloud storages. (c) By referencing the identifier, users can select and playback a Multisensory Memory, enabling the recorded experience to be shared with third person.

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Abstract

Conventional digital devices for sharing memorable experiences focus almost exclusively on audio-visual content, and no platform currently supports the Recording, Storing, and Playback of multi-sensory memories that include haptics. In this study, we develop an end-to-end platform and a compact hardware embodiment—the

Experience Sharing Box—to prototype the full pipeline and demonstrate the feasibility of sharing multisensory memories using the proposed system.

CCS Concepts

• **Human-centered computing** → **User interface management systems**.

Keywords

spatial computing, haptic feedback, experience sharing

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1 Introduction

Memory sharing began with murals and manuscripts, advanced through photography and print, and today relies on digital cameras that let people post experiences online. Recording, storing, and replaying such media transcends time and place, reinforces family bonds, preserves culture, and supports tele-collaboration, rehabilitation, education, and training. Yet current media remain largely visual–auditory, lacking touch and proprioception. Recent haptic displays and spatial-computing techniques can now capture force, vibration, and other haptic cues together with sight and sound.

Real-time spatial-haptics systems such as HaptoClone [Makino et al. 2015] demonstrate mid-air touch between remote users, but they provide no means to record or archive the experience. Single-modality tools—FEELTECH Wear [Umehara et al. 2024] and HaptoRoom [Tsujita et al. 2024]—augment mixed-reality interaction with local haptics, yet still lack an integrated pipeline that records, stores, and replays synchronized audio-visual-haptic data or connects heterogeneous devices.

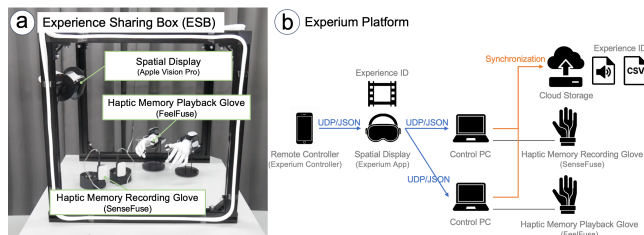


Figure 2: ESB and Experium Platform Architecture

2 Experience Sharing Box

Figure 2 shows the ESB hardware and Experium Platform for synchronized recording, storing, and playback. SenseFuse employs 16-channel FMG (Force Myography) sensors on each forearm and two fingertip contact-mic channels to capture muscle activity and vibrations. FeelFuse then delivers the signals through four rotational skin-stretch actuators on each forearm, linearly mapped to

muscle intensity, while fingertip vibrations are reproduced by two vibrotactile actuators per hand.

Fixing the Spatial Display inside the ESB lets the world coordinates captured in spatial video align exactly with those used during playback, eliminating drift and preserving realism. Haptic Memory Recording Gloves are worn during capture, and Haptic Memory Playback Gloves during replay. Devices are registered in the Experium app; time-stamped JSON commands sent over UDP via the control PC synchronise all start/stop events. Captured streams—spatial video, haptics, and FMG—are tied to a unique Experience ID and stored across local and cloud repositories. For replay, the app retrieves data by ID, starts the video, and sends matching JSON/UDP commands to the playback devices. When gaze input proves awkward, an external Remote Controller can issue the same commands while keeping recording and playback in sync.

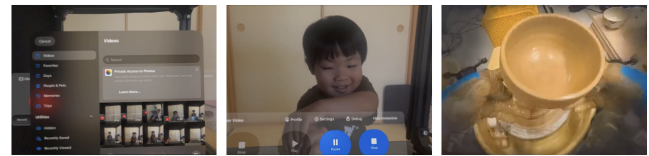


Figure 3: Select and Playback from multisensory Memories

3 Use Scenarios

ESB is designed to support a wide range of experience-sharing applications. As shown in Figure 3, the system enables the capture and playback of everyday activities, such as arm wrestling from a first-person view, as well as culturally significant practices like traditional pottery. Beyond daily family interactions—for example, allowing distant relatives to sustain embodied connection through shared multisensory experiences—ESB also supports the preservation and transmission of craft knowledge, enabling cultural practices to be recorded, reproduced, and shared.

In summary, ESB demonstrates the potential of recording, storing, and replaying multisensory memories across time and space to sustain embodied experiences.

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